



SAVEETHA SCHOOL OF ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES



Exp : 15 Write the python program to implement Decision Tree

Aim:

To write a python program to implement Decision Tree

Algorithm:

Step1: Import the required machine learning libraries and dataset.

Step2: Split the dataset into features and target labels.

Step3: Divide the data into training and testing subsets.

Step4: Initialize the Decision Tree classifier using an entropy or Gini impurity criterion.

Step5: Train the model by fitting it with the training dataset.

Step6: Predict the outcomes for the test dataset and evaluate the accuracy of the classifier.

Program:

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score

def build_decision_tree():
    data = load_iris()
    X = data.data
    y = data.target

    X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)

    classifier = DecisionTreeClassifier(criterion='entropy', random_state=42)
    classifier.fit(X_train, y_train)

    predictions = classifier.predict(X_test)
```

```

accuracy = accuracy_score(y_test, predictions)

return predictions, y_test, accuracy

if __name__ == '__main__':
    predicted_labels, actual_labels, model_accuracy = build_decision_tree()

    print("Predicted Labels:", predicted_labels)
    print("Actual Labels :", actual_labels)
    print("Model Accuracy :", round(model_accuracy * 100, 2), "%")

```

OUTPUT:

```

Predicted Labels: [1 0 2 1 1 0 1 2 1 1 1 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0 0
0 1 0 0 2 1
0 0 0 2 1 1 0 0]
Actual Labels    : [1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0 0
0 1 0 0 2 1
0 0 0 2 1 1 0 0]
Model Accuracy   : 97.78 %

```

Result: Thus, the program was executed and found that the output was found to be correct.